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Unsupervised Pangenomic Analysis of Paenibacillus polymyxa: Resolving Taxonomic Divergence and Pathogenic Risk

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1. Introduction

Paenibacillus polymyxa is a frequent inhabitant of the plant rhizosphere, where various strains act as beneficial plant growth-promoting bacteria by solubilizing phosphate, scavenging iron, secreting protective antibiotics, and, in specific genetic lineages, fixing atmospheric nitrogen [1]. While researchers are investigating functional variability and taxonomic differences among divergent strains, traditional taxonomic classification methods relying on highly conserved core genes, such as 16S rRNA, fail to accurately differentiate these divergent strains due to high sequence similarity (>98%), necessitating whole-genome analyses [1], [2].

This bioinformatic exploratory study was inspired by the exploratory questions proposed by Professor Melissa Chen from the Department of Biological Sciences at Simon Fraser University. Because empirical phenotypic and pathogenicity data are absent for most publicly sequenced draft genomes, traditional supervised classification approaches are ineffective, necessitating the translation of raw genomic data into a structured mathematical format suitable for unsupervised machine learning.

The primary objective of this study was to architect an unsupervised analytical pipeline to resolve ambiguous species boundaries, isolate unbalanced subpopulations, evaluate alternative partitioning models, and automate targeted functional profiling within the *P. polymyxa* species group. To achieve this, the study investigates three core biological objectives: establishing a baseline of structural separation based on global genomic variance, categorizing the cohort into distinct species groups, and isolating the genetic markers driving strain divergence. Furthermore, the framework incorporates an automated functional screening module to detect virulence factors and antibiotic resistance genes, enabling the identification of high-risk strains via a normalized Target Functional Index.

2. Methodology

2.1 Data Acquisition and Preprocessing

The 62 *P. polymyxa* strains analyzed in this project were identified using the finalized RefSeq assembly accession numbers from Supplementary File S4 of the Maggi et al. [2] study. These specific genomic assemblies and their annotations were downloaded from the NCBI database.

Draft genomes provided in raw GFF3 format were processed using a custom parsing module located in “src/matrix_generator.py”. To optimize the biological signal-to-noise ratio, the parser exclusively targeted lines annotated as Coding Sequences (CDS), explicitly filtering out structural RNAs and hypothetical proteins to prevent unwarranted feature bloat.

Functional features were extracted from the GFF3 files, prioritizing gene symbols (e.g., gene=paaA) or functional protein descriptions (e.g., product=kinase) to retain annotated metabolic enzymes. All records designated as hypothetical proteins were excluded from the final

dataset to prevent analytical noise generated by the false alignment of unverified, unrelated sequences across strains.

Cleaned and unique genetic inventories derived from all 62 strains were pooled to construct a master vocabulary index. This index was compiled into a binary presence/absence matrix yielding a dimension of 62 rows corresponding to individual strains and 4,460 columns corresponding to unique functional genes, where a value of 1 indicated feature presence and 0 indicated absence.

2.2 Pangenome Stratification

To characterize the pangenome architecture, genes from the binary matrix were stratified by strain frequency into three functional layers: Core (≥95%), Shell (15–95%), and Cloud (<15%).

2.3 Dimensionality Reduction and Clustering

To address distinct biological questions regarding structural variance, population partitioning, and genetic drivers, the analytical pipeline deploys an integrated sequence of classical machine learning models. To resolve global structural separation, Principal Component Analysis (PCA) is applied to perform dimensionality reduction on the binary pangenome feature matrix. By maximizing global variance, PCA condenses the matrix into a two-dimensional coordinate space, enabling the visual resolution of structural variance and the extraction of the top gene loading drivers.

HCA is then implemented on the binary matrix using complete linkage and a Jaccard distance metric, mathematically partitioning the strains into discrete evolutionary sub-lineages to provide a hierarchical dendrogram representation of evolutionary branching. As an alternative method to partition populations, K-Means (k=4) is evaluated directly on the PCA coordinates to compare against the HCA results.

2.4 Biomarker Identification and Functional Risk Screening

To identify specific genetic markers and functional profiles driving separation, an automated biomarker module calculates the absolute mathematical loading weights of every gene across Principal Component 1 (PC1) and Principal Component 2 (PC2) to extract the top individual gene drivers underlying divergence.

Furthermore, to enable preliminary discovery and high-risk strain screening, a targeted functional profiling module scans gene column labels using case-insensitive regular expression matching against a customizable list of target keywords. These specific target keywords (e.g., *paenilan*, *beta_lactamase*, *hemolysin*) were selected based on recent literature to directly screen for known bacteriocins, widespread antibiotic resistance, and general virulence factors. Strains are subsequently assigned a normalized Target Functional Index ranging from 0.0 to 1.0, based on matching risk-associated gene counts relative to the dataset maximum.

3. Results and Discussion

3.1 Core, Shell, and Cloud Distribution

Analysis of the global pangenome architecture across the 62-genome cohort revealed a total vocabulary of 4,460 unique functional features, stratified into a core genome of 1,885 genes present in at least 95% of strains (42.3%), a shell genome of 963 genes present in 15% to under 95% of strains (21.6%), and a cloud genome of 1,612 genes present in under 15% of strains (36.1%) **(Figure 1)**.

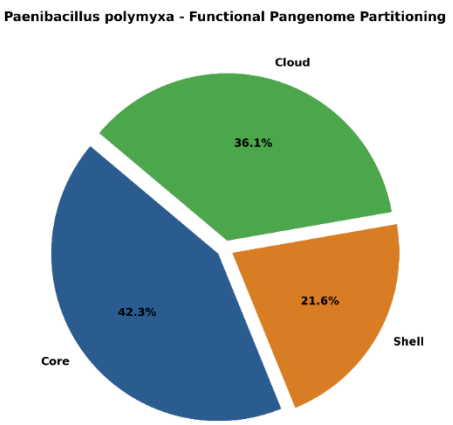


Figure 1: Distribution of the *P. polymyxa* pangenome feature space into Core (≥95%), Shell (15% - 95%), and Cloud (<15%) genetic. layers.

3.2 Taxonomic Divergence and Clustering

Principal Component 1 (PC1) captured 17.5% and Principal Component 2 (PC2) captured 10.2% of global structural variance. The HCA model specifically generates a dendrogram plot to provide a hierarchical representation of this evolutionary branching **(Figure 2)**. When executing K-Means (k=4) on the 2D PCA coordinate space and HCA directly on the binary presence/absence matrix, both clustering methods produced slightly different structural groupings **(Figure 3)**. This variation is expected given the distinct underlying data inputs and mathematical distance formulations (such as the Jaccard distance metric used for HCA) governing their partitioning mechanics.

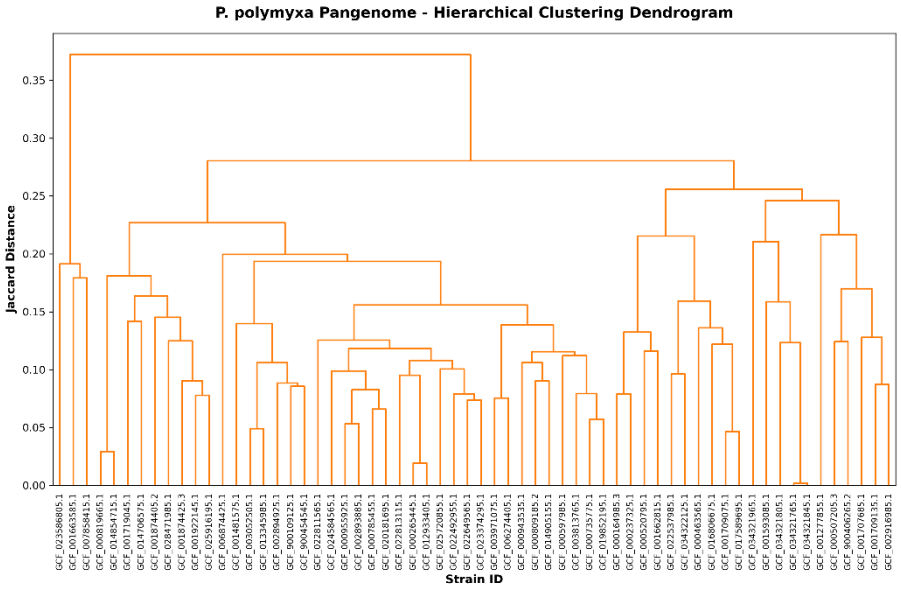


Figure 2: Evolutionary branching and structural variance of *P. polymyxa* strains, computed using complete linkage and a Jaccard distance metric.

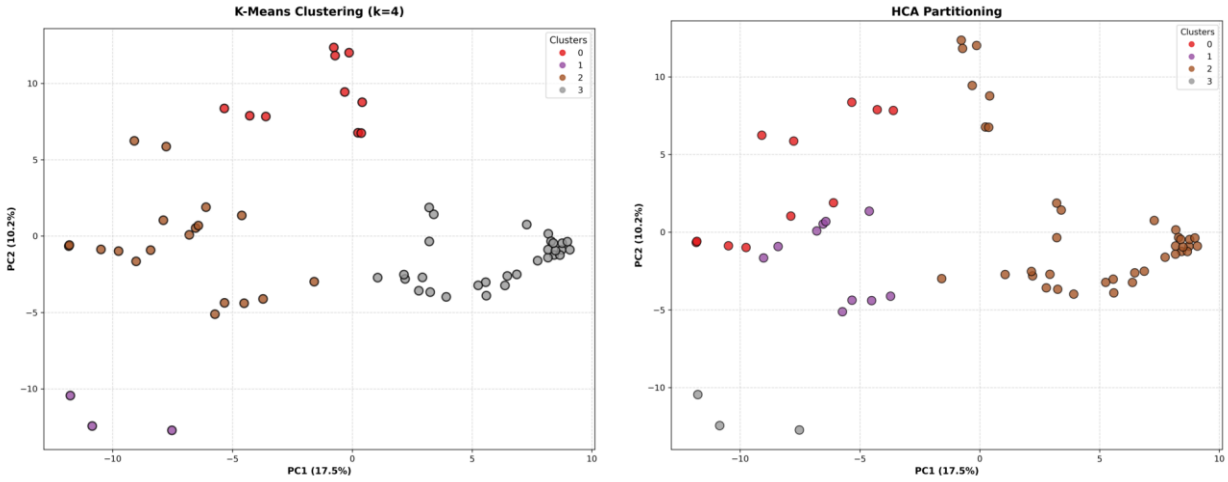


Figure 3: Side-by-side comparison of strain clustering using K-Means (k=4) overlay (left) and Hierarchical Cluster Analysis (HCA) partitioning labels (right).

Despite partitioning differences, both clustering methods successfully isolated the same three-strain grouping within the outgroups (GCF_001663585.1, GCF_007858415.1, and GCF_023586805.1). However, comparing these results to the baseline established by Maggi et al. [2] reveals notable taxonomic misalignments. While our pipeline correctly places *P. polymyxa* strains ND24 (GCF_001663585.1) and ZF197 (GCF_007858415.1) in the outgroup, it misclassifies *Paenibacillus ottowii* (GCF_006874425.1)**,** designated by Maggi et al. [2] as the primary outgroup species, into the same cluster as ATCC 842 (GCF_022811565.1). ATCC 842 serves as the type strain for *P. polymyxa*, acting as the baseline taxonomic reference point to define the true lineage in recent comparative genomic studies [1], [2]. Furthermore, our pipeline's outgroup includes *P. polymyxa* O 2.11 (GCF_023586805.1), which groups alongside with ATCC 842 in Maggi et al. [2].

These results diverge from Maggi et al. [2] due to fundamental differences in matrix preparation. While the reference study clustered genes by analyzing actual amino-acid sequence similarity, our pipeline relies on matching the text labels of existing RefSeq annotations to build a filtered presence/absence matrix. Because our approach evaluates functional text repertoire rather than biological sequence orthology, these variations in cluster assignments are expected.

3.3 Functional Biomarker Identification

To address the third objective regarding the specific genetic markers, post-hoc loading extraction revealed that spatial separation is driven by specific genetic markers; please see the top loading results for PC1 and PC2 in **Table 1** and **Table 2**. These top PC1 and PC2 drivers highlight the underlying genomic variance and specific functional features mathematically driving structural separation among the strains.

3.4 Pathogenicity Risk Assessment

Targeted functional profiling using the tailored high-risk keywords evaluated the cohort for virulence and resistance potential. The detailed pipeline results highlighting the top 10 highest-risk strains, including their calculated Target Functional Indices and corresponding cluster classifications, are provided in **Table 3**.

4. Conclusion

This unsupervised approach resolves the missing-label dilemma by enabling researchers to process novel draft genomes through this analytical pipeline to simultaneously map structural groupings, infer evolutionary lineage via hierarchical clustering, and evaluate pathogenic risk without immediate, expensive wet-lab plant trials. Because the flexible keyword-based module allows investigators to supply specialized terms based on domain expertise, the pipeline serves as a valuable tool for the preliminary discovery and exploration phase. While results remain dependent on the chosen vocabulary, utilizing targeted functional terms enables investigators to rapidly flag and prioritize strains of interest for downstream experimental validation.

5. Limitations & Future Work

5.1 Methodological Limitations

Regarding methodological limitations, the sparse binary presence/absence matrix ignores localized nucleotide variations such as Single Nucleotide Polymorphisms (SNPs) or promoter mutations. If a strain possesses a mutated, non-functional toxin gene, the matrix still registers it as present, potentially generating false-positive risk profiles. Furthermore, features extracted from GFF3 files rely on computational sequence homology predictions rather than wet-lab verifications, and PCA is strictly linear, struggling to capture complex multi-gene epistatic interactions without independent variance evaluations.

5.2 Future Directions

Future research directions include implementing non-linear dimensionality reduction techniques, such as Deep PyTorch Autoencoders or t-SNE, to map epistatic gene interactions into a continuous latent space, as well as leveraging unsupervised structural clusters and risk metrics to train lightweight supervised classifiers (such as K-Nearest Neighbors or Support Vector Machines), transforming the descriptive pipeline into a rapid clinical or agricultural diagnostic tool.

6. Project Experience Summary: Unsupervised Genomic Surveillance Pipeline

- **Resolved ambiguous taxonomic boundaries and identified evolutionary sub-lineages** within a cohort of 62 *P. polymyxa* draft genomes by developing an unsupervised machine learning pipeline in Python.
- **Translated raw, unstructured biological data into a structured mathematical format** by building a custom data parser using Pandas to extract Coding Sequences (CDS) from GFF3 files, generating a 62 x 4,460 binary presence/absence feature matrix while successfully filtering out biological noise like hypothetical proteins.
- **Mapped global genomic structural variance and visually partitioned populations** by engineering a dimensionality reduction and clustering pipeline utilizing Principal Component Analysis (PCA), K-Means, and Hierarchical Agglomerative Clustering (HCA) models via Scikit-Learn.
- **Extracted specific genetic drivers underlying strain divergence** by computationally calculating and isolating the top absolute loading weights of genes across principal components.
- **Eliminated the immediate need for expensive wet-lab plant trials** by engineering an targeted functional profiling module that rapidly screens strains using regex keyword matching (e.g., detecting toxins or antibiotic resistance genes) to assign a normalized Pathogenicity Risk Index for downstream experimental prioritization.

Key Technologies & Skills: Python, Pandas, Scikit-Learn, Machine Learning (PCA, K-Means, HCA), Data Cleaning & Preprocessing, Regular Expressions (Regex), Data Visualization (Matplotlib, Dendrograms), Bioinformatics.

References

[1] A. Wallner, L. Antonielli, O. Mesguida, P. Rey, and S. Compant, “Genomic diversity in Paenibacillus polymyxa: unveiling distinct species groups and functional variability,” *BMC Genomics*, vol. 25, no. 1, p. 720, Jul. 2024, doi: 10.1186/s12864-024-10610-w.

[2] F. Maggi *et al.*, “Pangenome analysis of *Paenibacillus polymyxa* strains reveals the existence of multiple and functionally distinct *Paenibacillus* species,” *Appl. Environ. Microbiol.*, vol. 90, no. 11, pp. e01740-24, Nov. 2024, doi: 10.1128/aem.01740-24.

Appendix

GENE_NAME	PC1_LOADING	PC2_LOADING
LOG_FAMILY_PROTEIN	0.063024	0.018968
TIGR00730_FAMILY_ROSSMAN_FOLD_PROTEIN	-0.063024	-0.018968
Y_FAMILY_DNA_POLYMERASE	0.060860	-0.021861
PLP_DEPENDENT_AMINOTRANSFERASE_FAMILY_PROTEIN	-0.060118	-0.017066
YLMH_FAMILY_RNA_BINDING_PROTEIN	0.059843	0.024608
RIBOSOMAL_MATURATION_YJGA_FAMILY_PROTEIN	0.059843	0.024608
DUF2809_DOMAIN_CONTAINING_PROTEIN	-0.059843	-0.024608
AMINOTRANSFERASE_LIKE_DOMAIN_CONTAINING_PROTEIN	0.059843	0.024608
RHAMNULOKINASE	0.058763	-0.038551
POLY_ETHYLENE_TEREPHTHALATE_HYDROLASE_FAMILY_PROTEIN	0.058712	-0.027182

Table 1: Top 10 genetic biomarkers driving global structural separation along Principal Component 1 (PC1).

GENE_NAME	PC1_LOADING	PC2_LOADING
ZINC_DEPENDENT_ALCOHOL_DEHYDROGENASE_FAMILY_PROTEIN	-0.039772	0.065043
FRUCTOSELYSINE_6_KINASE	0.022649	-0.064880
DUF4395_DOMAIN_CONTAINING_PROTEIN	-0.035741	0.064699
SUBSTRATE_BINDING_PERIPLASMIC_PROTEIN	-0.035741	0.064699
POTASSIUM_TRANSPORTING_ATPASE_SUBUNIT_F	-0.043001	0.064033
SRPBCC_FAMILY_PROTEIN	-0.027152	0.063840
MONOOXYGENASE	0.017660	-0.063658
ANT_6	0.046754	-0.063500
RIC	-0.046754	0.063500
DUF542_DOMAIN_CONTAINING_PROTEIN	0.046754	-0.063500

Table 2: Top 10 genetic biomarkers driving global structural separation along Principal Component 2 (PC2).

	TARGET_ASSOCIATED_GENE_COUNT	TARGET_FUNCTIONAL_INDEX	PC1	PC2	HCA_CLUSTER	KMEANS_CLUSTER
GCF_001707685.1	29	1.000	-5.337558	8.361183	0	0
GCF_006274405.1	28	0.966	8.169031	-1.410821	2	3
GCF_003971075.1	27	0.931	3.254390	-3.670401	2	3
GCF_001709135.1	27	0.931	-4.276718	7.884708	0	0
GCF_001709075.1	26	0.897	-4.609549	1.352788	1	2
GCF_017589695.1	26	0.897	-6.421629	0.688195	1	2
GCF_034322125.1	26	0.897	-9.019261	-1.654397	1	2
GCF_022811565.1	25	0.862	8.298529	-0.338667	2	3
GCF_025916195.1	25	0.862	-0.714751	11.823193	2	0
GCF_028471985.1	25	0.862	-0.312714	9.440202	2	0

Table 3: Top 10 highest-risk *P. polymyxa* strains prioritized by Target Functional Index and associated target gene counts.